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RESEARCH ARTICLE

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Validation of de novo designed water-soluble and transmembrane β-barrels by in silico folding and melting

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Abstract

In silico validation of *de novo* designed proteins with deep learning (DL)-based structure prediction algorithms has become mainstream. However, formal evidence of the relationship between a high-quality predicted model and the chance of experimental success is lacking. We used experimentally characterized *de novo* water-soluble and transmembrane β-barrel designs to show that AlphaFold2 and ESMFold excel at different tasks. ESMFold can efficiently identify designs generated based on high-quality (designable) backbones. However, only AlphaFold2 can predict which sequences have the best chance of experimentally folding among similar designs. We show that ESMFold can generate high-quality structures from just a few predicted contacts and introduce a new approach based on incremental perturbation of the prediction ("in silico melting"), which can reveal differences in the presence of favorable contacts between designs. This study provides a new insight on DL-based structure prediction models explainability and on how they could be leveraged for the design of increasingly complex proteins; in particular membrane proteins which have historically lacked basic *in silico* validation tools.

KEYWORDS
AlphaFold2, de novo design, ESMFold, membrane protein, structure prediction, β-Sheet

1 | INTRODUCTION

De novo protein design (Huang et al., 2016; Korendovych & DeGrado, 2020) is the generation of protein sequences and structure from scratch with no detectable homology to natural proteins of similar structure (Brunette et al., 2015; Kuhlman et al., 2003; Watson et al., 2023) and functions (Dou et al., 2018; Joh et al., 2014). Most *de novo* protein design pipelines build on a common paradigm: (i) sequence-free protein backbones are assembled with selected structure properties and (ii) used to guide combinatorial design of sequences (Huang et al., 2016; Kuhlman & Bradley, 2019). The sequences are then experimentally evaluated for expression and for folding. The percentage of designs that meet the experimental criteria for success is variable. Entire batches of designs can fail to express or fold because they were generated based on non-designable protein

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backbones. In-silico-generated protein backbones are considered "designable" if they incorporate key structural features of natural proteins (Dou et al., 2018; Koga et al., 2021). Features that define designable backbones include the chirality and the structure of the connections between secondary structures (Koga et al., 2012; Marcos et al., 2018), secondary structure packing (Dou et al., 2018; Koga et al., 2021; Lin et al., 2015), and the presence of torsional irregularities (e.g., kinks or bulges; Dou et al., 2018). Correct definition of these elements during the first steps of the design process reduces strain and steric clashes in the backbone. In practice, however, there are few explicit guidelines to help generate designable backbones, and the task requires significant expertise. Even when designable backbones are available, the *de novo* design of certain classes of proteins—such as complex β-sheet folds (Chidyausiku et al., 2022; Dou et al., 2018; Rocklin et al., 2017) or membrane proteins (MPs) (Duran & Meiler, 2018; Vorobieva, 2021)—can be challenging. The experimental success rate of de novo design (the number of designs from one batch that form a major soluble species with the expected secondary structure and oligomeric state) can change by an order of magnitude with the complexity of the fold (Rocklin et al., 2017)—α-helical folds typically achieving higher success rate (up to 50%) (Huang et al., 2014) than β-sheet folds (can be as low as 5%–10%) (Marcos et al., 2018).

To improve the success rate, the designs are often filtered and ranked for experimental validation by predicting *in silico* the most likely 3D structure adopted by the linear chain of amino acids (*ab initio*) to probe the sequence-structure compatibility (Bradley et al., 2005; Koga et al., 2012). However, classic *ab initio* predictions (e.g., Rosetta; Leman et al., 2020) often lead to inconclusive results as the complexity of the protein fold increases with size, contact order, and β-sheet content (Pan & Kortemme, 2021). For MPs, the implementation of an implicit membrane environment (Alford et al., 2020; Barth et al., 2007) makes it possible to screen only relatively simple designs such as self-assembling transmembrane α-helical peptides (Elazar et al., 2022; Lu et al., 2018; Scott et al., 2021). As a result, MP design still lacks many *in silico* validation tools available for water-soluble proteins (Vorobieva, 2021).

Recently, deep learning (DL) models such as AlphaFold2 (Jumper et al., 2021), RoseTTaFold (Baek et al., 2021), and language models (LMs) (e.g ESMFold [Lin et al., 2023], OmegaFold [Wu et al., 2022], or RGN2 [Chowdhury et al., 2022]) have out-performed classic *ab initio* folding simulations. AlphaFold2, RoseTTAFold, and ESMFold networks have also been inverted to design new protein sequences (Anishchenko et al., 2021; Goverde et al., 2023; Verkuil et al., 2022). We sought to evaluate the opportunities and limitations of using AlphaFold2 and protein LMs—exemplified here by ESMFold—for *in silico* validation and ranking of *de novo* designed proteins. We assembled three datasets of experimentally characterized synthetic proteins considered as "challenging targets" for classic *ab initio* structure prediction: (i) sequences generated based on un-designable backbones, (ii) β-barrel proteins designed to fold in water, and (iii) in lipid membranes. We predicted the structures of the proteins in the datasets using standard and modified AlphaFold2 and ESMFold pipelines to reveal differences between the sequences that successfully passed or failed experimental validation.

2 | RESULTS

2.1 | Designable and un-designable backbones

We first sought to assess AlphaFold2 and ESMFold capacity to distinguish designs generated based on designable and un-designable backbones. We assembled two datasets of 83 sequences designed to fold into β-barrels (Dou et al., 2018) (see methods, Table S1), a fold formed from one unique β-sheet that curves on itself to close into a tube-shaped structure. A set of 42 designs incorporated specific structure features (kinks) that helped reduce strain in the curved β-sheet, achieving a success rate of about 35% ("designable" dataset). The second dataset contained 41 designs generated with perfectly regular β-strands lacking kinks ("un-designable" sub-group) (Murzin et al., 1994; Novotný et al., 1984). None of these sequences folded into the expected β-barrel structure. Instead, they failed to express or were insoluble. All 83 designs (designable and un-designable) passed the classic *ab initio* folding test and showed a funnel-shaped *in silico* folding landscape converging to the design model (Figure S1). Hence, classic *ab initio* structure prediction was unable to distinguish designs from the two datasets. We modeled the 3D structures with standard AlphaFold2 and ESMFold pipelines as described in the methods section and used two metrics to assess model quality: (1) the confidence assigned to the prediction (predicted local distance difference test, pLDDT) and (2) the root mean square deviation (RMSD) of the predicted structure and the design model.

Both ESMFold and AlphaFold2 predicted the structures of *de novo* designed proteins from a single sequence (without a multiple sequence alignment), which clearly clustered into distinct "designable" and "un-designable" populations (Figure 1a,b). "Designable" sequences yielded, in average, more similar structures to the expected design models and were attributed higher

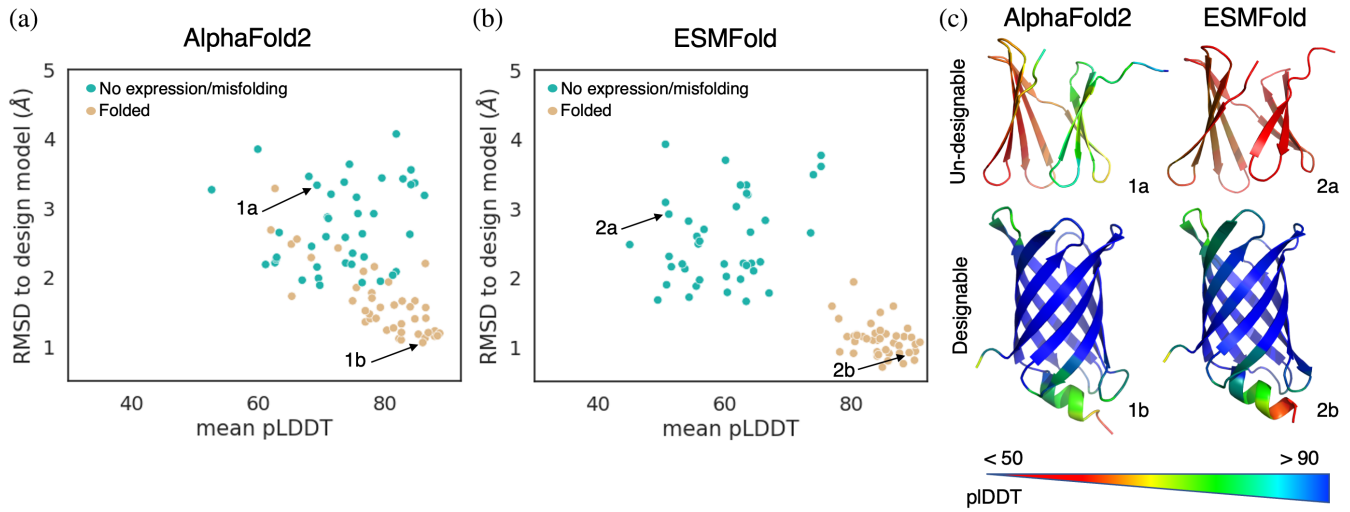


FIGURE 1 Predicting backbone designability. AlphaFold2 (a) and ESMFold (b) predictions for sequences generated based on designable backbones sample higher pLDDT and lower RMSD values than the "un-designable" dataset. (c) Example AlphaFold2 and ESMFold models colored by pLDDT.

confidence (Figure 1c). EMSFold performed significantly better on the dataset tested here; all "designable" β-barrel models had a pLDDT score higher than 0.75 and a RMSD lower than 2.0 Å.

2.2 | Folded and misfolded designed sequences

Even with designable backbones as starting points, the number of successfully folding designs within a batch of designs can be low, despite relatively similar computed energies and sequences (40%–75% sequence identity). The determinants of success/failure remain elusive and likely more subtle than those underlying designable backbones. We sought to determine whether these differences were captured by AlphaFold2 and ESMFold.

2.3 | Overview of the datasets

We assembled datasets of *de novo* designed β-sheet proteins folding in water and in lipid membranes and labeled according to their experimental behavior: expressed, soluble (in water or detergent micelles) and successfully folded (see experimental validation criteria, Table S1). All-β-sheet folds were selected here because they represent particularly challenging targets for classic *ab inito* folding simulations due to the presence of long-range (and potentially ambiguous) interactions between β-strands. Despite sharing similar structure properties, water-soluble and transmembrane β-sheet folds have very different sequence properties (Dhar et al., 2021).

The water-soluble dataset comprised 42 β-barrels generated based on designable backbones. Fifteen of these sequences were correctly folded (∼35% success rate) as attested by the presence of a detectable monomeric species with the expected β-sheet secondary structure. No difference between these 15 folded designs and the 27 "failed" (non-expressing, aggregated, or misfolded) designs was detected using Rosetta *ab initio* structure prediction. The transmembrane dataset is, to our knowledge, the largest published dataset of experimentally characterized synthetic MPs, composed of 115 sequences designed to fold into small eight-stranded transmembrane β-barrels (TMB) (Vorobieva et al., 2021). A major challenge for TMB design is the necessity to incorporate local sequence/structure frustration to modulate the folding/aggregation kinetics. All 25 designs in our dataset that did not incorporate frustrated residues (strong β-sheet designs) failed to express in *Escherichia coli* (Vorobieva et al., 2021). Among the 90 designs generated with local secondary structure frustration (weak β-sheet designs), 17 folded as expected into monomeric β-sheet proteins in detergent micelles (∼19% success rate) (Vorobieva et al., 2021). TMB designs were not evaluated *in silico* due to technical limitations of ab initio folding simulations (Alford et al., 2020; Alford et al., 2015).

2.3.1 | Sequence-structure compatibility assessment with AlphaFold2 folding

As previously observed (Dauparas et al., 2022; Lin et al., 2023; Wang et al., 2022; Wicky et al., 2022), AlphaFold2 could predict the structure of water-soluble designs

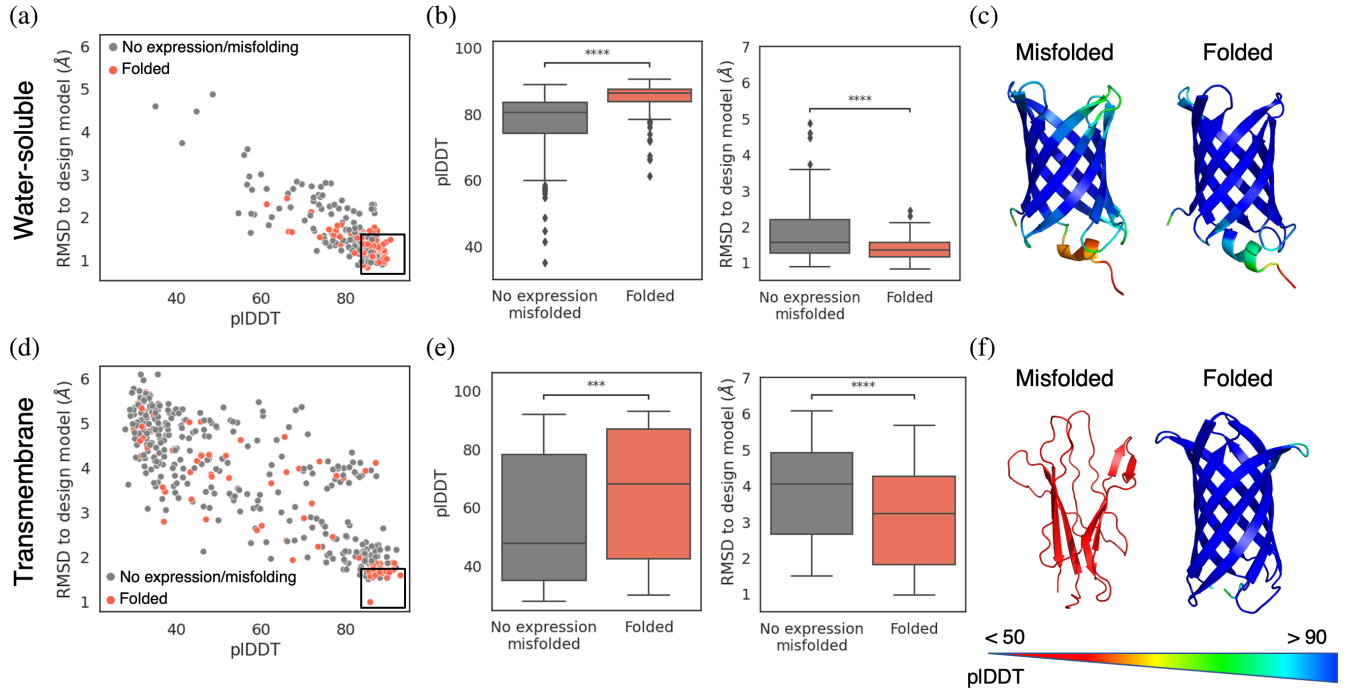


FIGURE 2 AlphaFold2 generates significantly different predictions for successfully folding and misfolding water-soluble (a–c) and transmembrane (d–f) β-barrel designs. Five models were generated per sequence; panel (a) shows 210 simulations for 42 water-soluble β-barrels; panel (d) shows 405 simulations for 81 TMBs.

from a single sequence. We found that experimentally folded and misfolded proteins sampled significantly different distributions of pLDDT and RMSD (p-values 10^{-11} and 10^{-5}, respectively) (Figure 2a,b). Folded designs were, on average, attributed higher pLDDT confidence and were more similar to their respective design models than the designs that misfolded (Figure 2b,c). Out of five generated models, successfully folded designs yielded significantly more high-quality models (pLDDT ≥0.85 and RMSD ≤1.75 Å) than misfolded designs (Figure S2a). Considering the lowest-RMSD model only (Figure S2b), the chance of picking a folded sequence (precision) was 89% when pLDDT was higher than 0.85 and RMSD lower than 1.75, and accuracy was 79%. The receiver operator characteristic (ROC) curves are shown in Figure S2c.

In contrast to water-soluble proteins, the capacity of AlphaFold2 to predict the structure of designed MPs from a single sequence was unknown (MPs represent less than 2% of structures used for AlphaFold2's training (White, 2009)). We evaluated it on our dataset of 115 *de novo* designed TMBs. Surprisingly, AlphaFold2 predictions with the best pLDDT and RMSD scores were obtained for designs that failed to express (p-value [pLDDT] $= 10^{-4}$; p-value [RMSD] $= 10^{-4}$) (Figure S3a,b). We reasoned that the "strong β-sheet" designs in our dataset (which largely failed to express) might also feature more distinct 3D structure signatures than the frustrated "weak β-sheet" designs and therefore yield higher quality models (Figure S3c). We decided to focus our analysis on the sequences that can be expressed and filtered-out "strong β-sheet" sequences based on empirical β-sheet (50%, RaptorX; Wang et al., 2016) and aggregation propensity cutoffs (1750, Tango; Fernandez-Escamilla et al., 2004) (Figure S3d). However, most of the remaining 81 TMBs were attributed low pLDDT and high RMSD scores and no difference was observed between the folded and misfolded designs (p-value [pLDDT] $= 0.99$; p-value [RMSD] $= 0.33$). We hypothesized that TMBs were challenging targets for single sequence AlphaFold2 because of the designed sequence-structure frustration, and that increasing the number of recycles through the Evoformer (Jumper et al., 2021) could help resolving weak 3D contacts. The predictions dramatically improved with 48 recycles (instead of three): successfully folded designs sampled significantly better distributions of pLDDT (p-value $= 10^{-4}$) and RMSD (p-value $= 2 \times 10^{-4}$) than the designs that misfolded (Figure 2d,f). AlphaFold2 generated at least one full β-barrel structure for 13 out of 16 folded designs, while misfolding sequences yielded mostly low confidence, unstructured models (Figure S2d, Figure 2f). Considering the lowest-RMSD model only (Figure S2e), the precision was 48% when pLDDT was higher than 0.85 and RMSD lower than 1.85 Å (accuracy 79%, ROC curves in

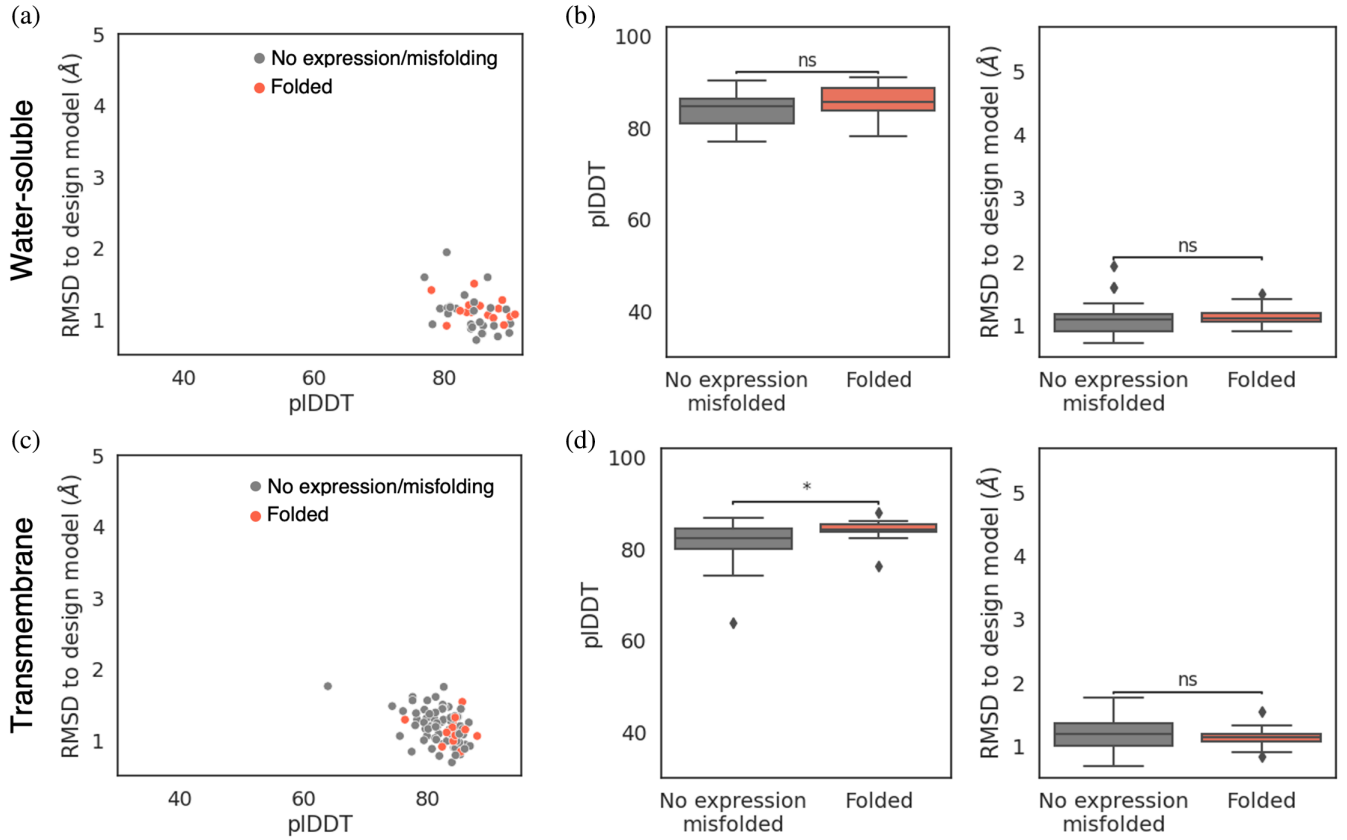


FIGURE 3 ESMFold generates high-confidence predictions closely resembling the design models for water-soluble (a,b) and transmembrane β-barrels (c,d). One model was generated for each of the 42 water-soluble β-barrels (a) and 90 "weak β-sheet" TMBs designed (c).

Figure S2f). This represents a 2.5-times increase in success rate by comparison to the previous "naive" design selection. Hence, a carefully designed AlphaFold2 screen taking into consideration the biophysical specificities of the fold of interest could improve the success of *de novo* protein design by two to three times.

2.3.2 | Sequence-structure compatibility determination with ESMFold

Protein LMs, such as ESM, have been developed to capture protein properties from a single sequence. Therefore, ESMFold is often considered as a suitable tool for the analysis of synthetic (*de novo* designed) and orphan protein sequences, for which few homologs can be found. The software comprises a language module (LM) predicting residue contacts, and a structure module (SM) generating 3D coordinates based on the input sequence and the predicted contact (Lin et al., 2023). Strikingly, ESMFold generated high pLDDT, low RMSD models for nearly all sequences tested here (Figure 3). It could predict the structures of *de novo* designed TMBs with comparable performance to water-soluble β-barrels and made almost no distinction between experimentally folded and misfolded designs (Figure 3b–d and ROC curves in Figure S4a–c). The predicted TMscore (pTM) score had only marginally higher predictive power over pLDDT (Figure S4).

2.3.3 | Beyond the known membrane protein topologies

To assess the capacity of AlphaFold2 and ESMFold to extrapolate beyond known designed and natural β-barrels structures, we tested a set of 40 (including seven successfully folded) sequences designed to fold into 12-stranded TMBs without sequence nor structure similarity homology to β-barrels in the Protein DataBank (Berhanu et al., 2023). Overall, the two softwares performed as previously observed. AlphaFold2 generated significantly better models for experimentally folded over misfolded sequences (Figure 4a,b), ESMFold generated high-confidence structures closely resembling the design models for almost all tested designs, irrespective of

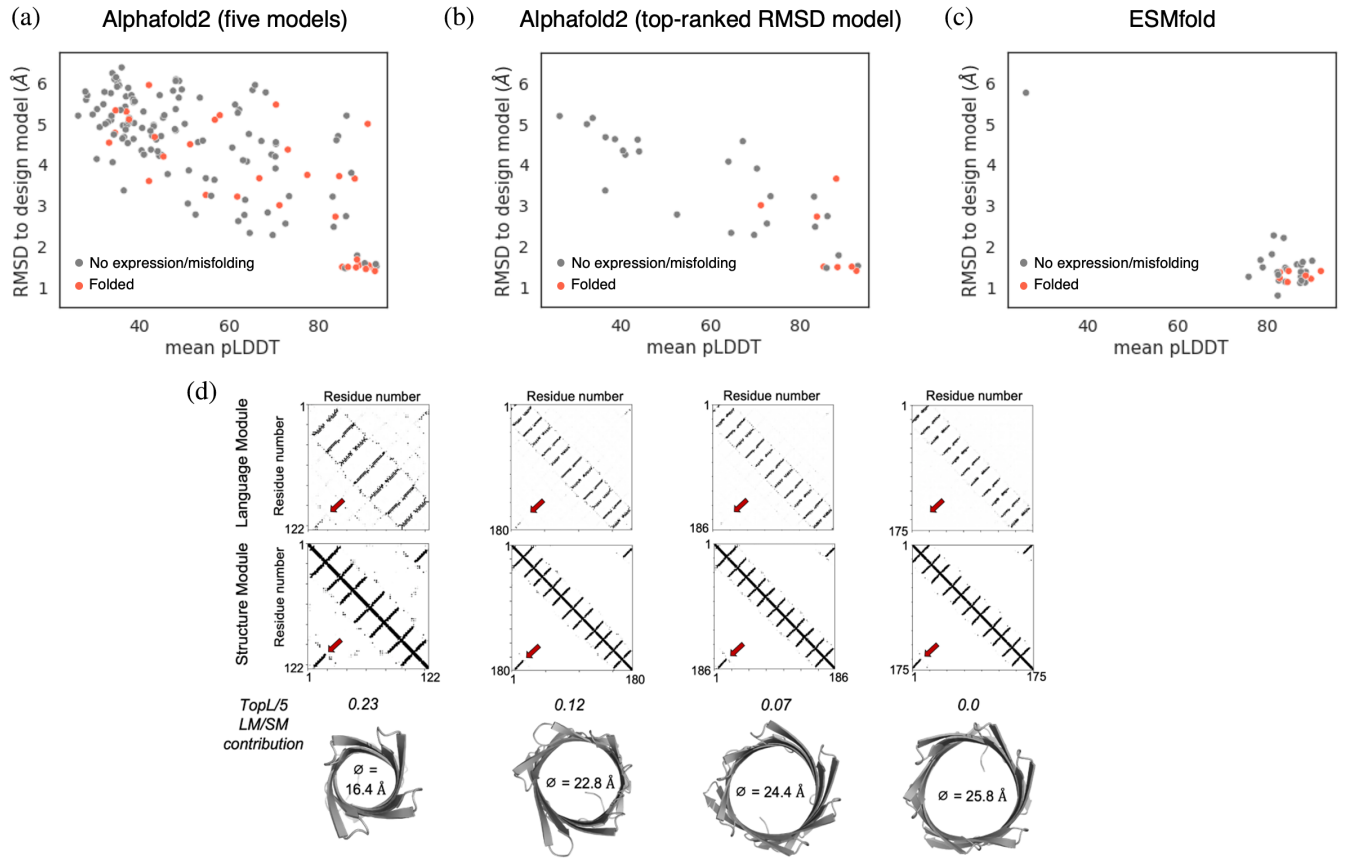


FIGURE 4 AlphaFold2 and ESMFold predictions for new-to-nature 12-stranded TMB designs. AlphaFold2 distinguishes successfully folding from failed designs, using all five models (a) or the top ranked (lowest RMSD) model (b). (c) ESMFold generates high-confidence, low RMSD, predictions for nearly all designs. (d) Fewer long-range contacts (TopL/5) are predicted by the LM as the diameter of the β-barrel increases. Yet, the TopL/5 contacts are correctly predicted by the SM.

experimental outcome (Figure 4c). Hence, AlphaFold2 and ESMFold were able to generate accurate single-sequence predictions of novel and complex MP designs. However, only AlphaFold2 was able to distinguish properly folded from misfolded proteins.

Although ESMFold generated accurate structures for most designs in the new dataset, two predictions differed from the design model at the level of the connectivity between the first and last β-strands. To investigate the origin of the discrepancy, we developed an ESMFold Colaboratory notebook that reports the residue contacts predicted by the LM and the SM. We used the notebook to predict the structures of TMBs of different sizes (diameters from 16.0 to 25.8 Å). We found that the capacity of the LM to detect long-range contacts (*TopL/5 LM/SM contribution*) decreased when the diameter of the β-barrel increased (Figure 4d). No contacts were predicted between the first and the last β-strands of the largest TMB. Yet, a closed β-barrel with the expected β-strands connectivity was generated by the SM. The observed cooperation between the LM and SM might explain the consistent results achieved by ESMFold on *de novo* designed sequences.

2.3.4 | *In silico* melting reveals difference between folding and misfolding designs

We further investigated the robustness of the ESMFold prediction by randomly perturbing the LM using a new approach dubbed *in silico* melting and implemented in Google Colaboratory. To systematically increase the noise level, we used the mask feature of the ESM LM (Lin et al., 2023) to mask a subset of randomly selected residues in the input sequence. Sixty-four predictions were sampled for each level of noise (percentage of the input sequence masked). We computed the average pTM score across the 64 models as well as a success rate for each noising step (see Methods). Four *de novo* designed proteins with a high-resolution structure in the PDB were initially analyzed. The predictions obtained for the α-helical bundle (Huang et al., 2014) and the α/β TOP7 (Kuhlman et al., 2003) designs were surprisingly robust; accurate structures were generated despite masking 90%–95% of the sequence. By contrast, the success rate dropped at around 55% noise for the water-soluble (Dou et al., 2018) and the transmembrane β-barrels (Vorobieva

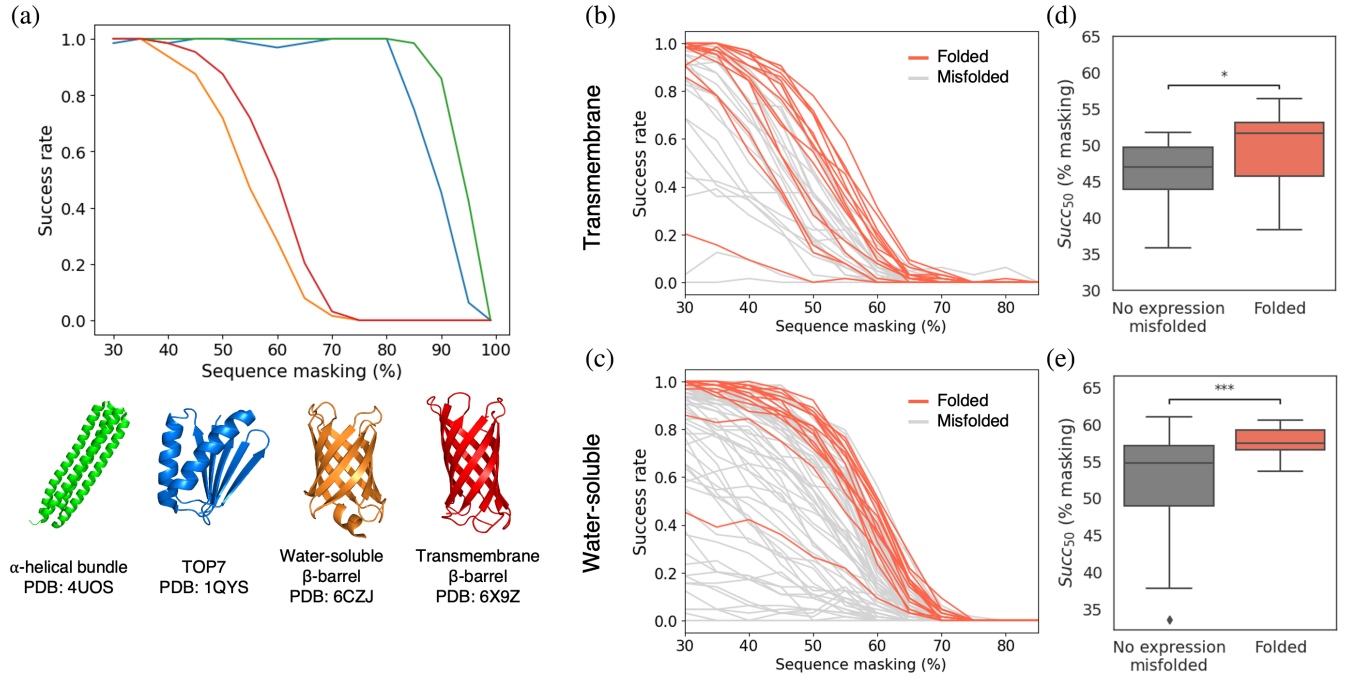


FIGURE 5 ESMFold sequence masking to assess design robustness. (a) ESMFold in silico melting curves for four de novo designed proteins. (b,c) Prediction robustness for water-soluble (b) and transmembrane (c) β-barrel designs. Folded designs are shown in red. (d,e) Distributions of sigmoidal inflection points for water-soluble (d) and transmembrane (e) designs (*p*-values of 2×10^{-2} and 4×10^{-4}, respectively).

et al., 2021) (Figure 5a). Hence, the SM can predict α-helices rich structures using just a few LM-predicted contacts but requires more contacts information for β-sheet proteins, likely because of limited capacity to interpret long-range contacts. Success rate followed sigmoid-shaped curves for the TMB and the water-soluble β-barrel, while pTM transitions were either sigmoid-shaped (TMB) or multi-modal in water-soluble β-barrel, indicating the presence of noise-resistant local secondary structure (Figure S5).

Focusing on the sigmoid-shaped success rate transitions, we compared designs with similar structures and sequences in the water-soluble and transmembrane β-barrel datasets. In both datasets, ESMFold predictions for successfully folded designs were more robust to sequence masking than the designs that failed to fold (Figure 5b,c). We fitted a sigmoidal model to the simulated melting curves to extract the inflection points of predictions ($Succ_{50}$, see Methods) and found a significant difference between the folded and misfolded designs (Figure 5d,e). Based on the fitted $Succ_{50}$ values, it is possible to find for both datasets an empirical cut-off value that yields accuracy and precision comparable to AlphaFold2 (calculated optimal cut-offs of 51.6 and 55.8 for water-soluble and transmembrane β-barrels, respectively) (ROC curves and confusion matrices in Figure S6). Most folded TMB designs were scored high by both AlphaFold2 and ESMFold melting (Figure S7). The RMSD, pLDDT, and $Succ_{50}$ metrics identified different false positives, suggesting that a combination of AlphaFold2 and ESMFold melting could result in a high-precision model (Figure S7, Table S2). By comparison, several true positive and false positive water-soluble β-barrels identified by AlphaFold2 had a low $Succ_{50}$ score, indicating higher precision but lower recall achieved by melting simulations (Figure S7, Table S3). A hypothetical scenario selecting the top 10 models based on $Succ_{50}$ would result in a success rate of 80% for the water-soluble barrels and 60% for the TMB designs.

3 | DISCUSSION

AlphaFold2 and ESMFold are routinely used to validate *de novo* designs and to compare the performance of design methods. Our results support previous observations that both models can accurately predict the structures of *de novo* designed proteins from a single sequence, although only ESMFold was specifically trained for that purpose. Remarkably, the structures of complex multi-span TMBs could be predicted, although they represent less than 0.5% of the structures in the PDB (White, 2009). Using designs generated on accurately assembled ("designable") β-barrel backbones and designs

incorporating non-natural backbone features ("un-designable"), we show that ESMFold could be used to rapidly and accurately assess backbone designability. It can predict the structure of a small protein in less than 1 min, which makes it suitable for high-throughput analysis of large sets of sequences at the early stage of *de novo* design. However, once appropriate backbone constraints have been defined, only AlphaFold2 could capture the subtle differences resulting in proper folding or misfolding. It has better discriminative power than classic *ab initio* prediction for β-sheet proteins—which are notoriously difficult to design and fold *in silico*—and can be used to prioritize designs for experimental validation. However, users should exercise caution since the prediction parameters might have to be adjusted to account for biophysical specificities of the fold (e.g., number of recycles for TMBs). We also propose a new ESMFold prediction perturbation strategy that could be used to reveal differences in intramolecular contacts robustness. Both methods have similar speeds and could correctly identify folded and misfolded designs in the water-soluble and TMB datasets with comparable accuracy and precision. Similar true positive but different false positive designs were identified, suggesting that the two models might be complementary. However, ESMFold melting misclassified "folded" water-soluble β-barrels more frequently than TMBs. This might be due to the difference in local (pTM) and global (success rate) robustness to noise observed for water-soluble designs, which makes the simulation more challenging to analyze and interpretate. Moreover, we note that $Succ_{50}$ is an abstract metric by comparison to RMSD and pLDDT and further tests will be necessary to propose general rules for the selection of a cut-off value.

Finally, this study suggests a potential risk in using ESMFold for protein design. In the examples presented here, even sparse contacts from the LM (as much as 90% of the sequence masked) can be captured and amplified by the SM to return the correct structure. While desirable for structure prediction, such behavior might be problematic for protein design with an inverted ESMFold network. The iterative optimization of the protein sequence is guided by a loss function that defines a gradient toward a high-quality predicted structure. If a high-quality prediction is achieved faster than strong contacts are introduced, the designs will contain minimal information to trigger the prediction of the correct structure but will be unlikely to fold experimentally. Moving forward, a strategy that randomly masks input positions during design may help fix this issue and design robust proteins with good "melting temperature," opening the door to creating increasingly complex water-soluble and MPs.

4 | MATERIALS AND METHODS

4.1 | AlphaFold modeling

AlphaFold's Colabfold version (AlphaFold2.ipynb) was used through this study (Mirdita et al., 2022). All predictions were made with the single_sequence MSA setting, no AMBER force-field relaxation, and no template. The number of recycles was set to 3 (for the designable/undesignable backbones test and for water-soluble β-barrels) or 48 (for transmembrane β-barrels). Five models were generated per input. The water-soluble β-barrels dataset was truncated to remove designs stabilized with disulfide bonds. Such designs usually differ only by a few mutations from a disulfide-free parent in the dataset but might have quite different experimental behavior. The transmembrane β-barrels dataset was truncated to exclude sequences with high β-sheet and/or aggregation propensities, as described in the main text.

4.2 | ESMFold modeling

ESMFold implemented in Colabfold (ESMFold_advanced.ipynb) was used through this study to generate structure predictions and visualize contacts from the language and the structure models. The number of recycles was set to 0.

4.3 | Evaluation metrics

The generated models were aligned to the coordinates of the original design model using TM-align (Zhang & Skolnick, 2005) to calculate the RMSD and the TM-score. To analyze the ESMFold predictions, the pLDDT scores of the backbone heavy atoms (N, CA, O, C, CB) were averaged over the entire sequence. After confirming the skewness of the distributions with a normality test, the Mann Whitney *U* test was used to compare the distributions of confidence scores (pLDDT) and RMSD between populations of designs. The results were confirmed using a permutation test. When the data were represented in a box plot, the center line shows the median; the box limits show the first and third quartiles; the whiskers show the minimum and maximum of the distribution; the points show outliers.

4.4 | In silico melting with ESMFold

The analysis was done using a custom ESMfold colab notebook (https://colab.research.google.com/github/voro

bieva/ColabFold/blob/main/beta/ESMFold_melting.ipynb). For each input sequence, perturbation was incrementally added to the prediction by applying a mask on randomly selected 30%–95% of residues in the sequence (5% increment). For each perturbation step, 64 models were generated with different and randomly applied masking patterns of non-sequential positions in the sequence. The success rate at each step was defined as the fraction of models out of 64 with a predicted TM-score (pTM) higher than 0.75, which is the threshold that enables discrimination between two β-barrels with the same number of strands but different β-strands connectivity. To quantitatively compare the robustness of the predictions, the success rate as a function of the level of perturbation was fitted to a sigmoidal curve to calculate the inflection point of the transition ($Succ_{50}$). Some simulations could not be fitted with a sigmoidal function because the predictions were significantly disrupted at the lowest tested noising rate (30%). These designs could not be included in the $Succ_{50}$ comparative analysis (Tables S2 and S3). To find the empirical $Succ_{50}$ cutoff, we selected the highest Youden's index (calculated as true positive rate minus false positive rate) for each possible $Succ_{50}$ threshold; at this $Succ_{50}$ cutoff we achieve the best balance between the TPR and FPR to maximize the performance of the classifier.

COMPUTE CODE STATEMENT

The code used to generate the predictions is available through the ColabFold repository: https://github.com/sokrypton/ColabFold and https://github.com/vorobieva/ColabFold//tree/main/beta/ESMFold_melting.ipynb to generate ESMFold melting predictions. The code for processing, visualizing and analyzing results is available at: https://github.com/vorobieva/validate_de_novo_designed_proteins.

AUTHOR CONTRIBUTIONS

Alvaro Martin Hermosilla: Investigation; visualization; formal analysis; data curation. **Carolin Berner:** Investigation; writing – review and editing; formal analysis. **Sergey Ovchinnikov:** Conceptualization; software; writing – review and editing; funding acquisition. **Anastassia A. Vorobieva:** Conceptualization; methodology; investigation; funding acquisition; writing – original draft; software; supervision.

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CONFLICT OF INTEREST STATEMENT

The authors declare no competing interests.

DATA AVAILABILITY STATEMENT

All source data are available with this paper. These data, as well as the models generated for this study, are available on Zenodo: https://zenodo.org/records/10968476.

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REFERENCES

Alford RF, Fleming PJ, Fleming KG, Gray JJ. Protein structure prediction and design in a biologically realistic implicit membrane. Biophys J. 2020;120:2042–55.

Alford RF, Koehler Leman J, Weitzner BD, Duran AM, Tilley DC, Alazar A, et al. An integrated framework advancing membrane protein modeling and design. PLoS Comput Biol. 2015;11: e1004398.

Anishchenko I, Pellock SJ, Chidyausiku TM, Ramelot TA, Ovchinnikov S, Hao J, et al. De novo protein design by deep network hallucination. Nature. 2021;600:547–52.

Baek M, DiMaio F, Anishchenko I, Dauparas J, Ovchinnikov S, et al. Accurate prediction of protein structures and interactions using a three-track neural network. Science. 2021;373:871–6.

Barth P, Schonbrun J, Baker D. Toward high-resolution prediction and design of transmembrane helical protein structures. Proc Natl Acad Sci U S A. 2007;104:15682–7.

Berhanu L, Majumder S, Müntener T, Whitehouse J, Berner C, Bera AK, et al. Sculpting conducting nanopore size and shape through de novo protein design. bioRxiv. 2023. https://doi.org/10.1101/2023.12.20.572500

Bradley P, Misura KMS, Baker D. Toward high-resolution de novo structure prediction for small proteins. Science. 2005;309: 1868–71.

Brunette T, Parmeggiani F, Huang PS, Bhabha G, Ekiert DC, Tsutakawa SE, et al. Exploring the repeat protein universe through computational protein design. Nature. 2015;528:580–4.

Chidyausiku TM, Mendes SR, Klima JC, Nadal M, Eckhard U, Roel-Touris J, et al. De novo design of immunoglobulin-like domains. Nat Commun. 2022;13:5661.

Chowdhury R, Bouatta N, Biswas S, Floristean C, Kharkar A, Roy K, et al. Single-sequence protein structure prediction using a language model and deep learning. Nat Biotechnol. 2022;40: 1617–23.

Dauparas J, Anishchenko I, Bennett N, Bai H, Ragotta RJ, Milles LF, et al. Robust deep learning-based protein sequence design using ProteinMPNN. Science. 2022;378:49–56.

Dhar R, Feehan R, Slusky JSG. Membrane barrels are taller, fatter, inside-out soluble barrels. J Phys Chem B. 2021;125:3622–8.

Dou J, Vorobieva AA, Sheffler W, Doyle LA, Park H, Bick MJ, et al. De novo design of a fluorescence-activating β-barrel. Nature. 2018;561:485–91.

Duran AM, Meiler J. Computational design of membrane proteins using RosettaMembrane. Protein Sci. 2018;27:341–55.

Elazar A, Chandler NJ, Davey AS, Weinstein JY, Nguyen JV, Trenker R, et al. De novo-designed transmembrane domains tune engineered receptor functions. Elife. 2022;11:e75660.

Fernandez-Escamilla A-M, Rousseau F, Schymkowitz J, Serrano L. Prediction of sequence-dependent and mutational effects on the aggregation of peptides and proteins. Nat Biotechnol. 2004; 22:1302–6.

Goverde CA, Wolf B, Khakzad H, Rosset S, Correia BE. De novo protein design by inversion of the AlphaFold structure prediction network. Protein Sci. 2023;32:e4653.

Huang P-S, Boyken SE, Baker D. The coming of age of *de novo* protein design. Nature. 2016;537:320–7.

Huang P-S, Oberdorfer G, Xu C, Pei XY, Nannenga BL, Rogers JM, et al. High thermodynamic stability of parametrically designed helical bundles. Science. 2014;346:481–5.

Joh NH, Wang T, Bhate MP, Acharya R, Wu Y, Grabe M, et al. De novo design of a transmembrane Zn^{2+}-transporting four-helix bundle. Science. 2014;346:1520–4.

Jumper J, Evans R, Pritzel A, Green T, Figurnov M, Ronneberger O, et al. Highly accurate protein structure prediction with AlphaFold. Nature. 2021;596:583–9.

Koga N, Koga R, Liu G, Castellanos J, Montelione GT, Baker D. Role of backbone strain in de novo design of complex α/β protein structures. Nat Commun. 2021;12:3921.

Koga N, Tatsumi-Koga R, Liu G, Xiao R, Acton TB, Montelione GT, et al. Principles for designing ideal protein structures. Nature. 2012;491:222–7.

Korendovych IV, DeGrado WF. *De novo* protein design, a retrospective. Q Rev Biophys. 2020;53:E3.

Kuhlman B, Dantas G, Ireton GC, Varani G, Stoddard BL, Baker D. Design of a novel globular protein fold with atomic-level accuracy. Science. 2003;302:1364–8.

Kuhlman B, Bradley P. Advances in protein structure prediction and design. Nat Rev Mol Cell Biol. 2019;20:681–97.

Leman JK, Weitzner BD, Lewis SM, Adolf-Bryfogle J, Alam N, Alford RF, et al. Macromolecular modeling and design in Rosetta: recent methods and frameworks. Nat Methods. 2020; 17:665–80.

Lin Z, Akin H, Rao R, Hie B, Zhu Z, Lu W, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. Science. 2023;379:1123–30.

Lin YR, Koga N, Tatsumi-Koga R, Liu G, Clouser AF, Montelione GT, et al. Control over overall shape and size in de novo designed proteins. Proc Natl Acad Sci USA. 2015;112: E5478–85.

Lu P, Min D, DiMaio F, Wei KY, Vahey MD, Boyken SE, et al. Accurate computational design of multipass transmembrane proteins. Science. 2018;359:1042–6.

Marcos E, Chidyausiku TM, McShan AC, Evangelidis T, Nerli S, Carter L, et al. De novo design of a non-local β-sheet protein with high stability and accuracy. Nat Struct Mol Biol. 2018;25: 1028–34.

Mirdita M, Schütze K, Moriwaki Y, Heo L, Ovchinnikov S, Steinegger M. ColabFold: making protein folding accessible to all. Nat Methods. 2022;19:679–82.

Murzin AG, Lesk AM, Chothia C. Principles determining the structure of β-sheet barrels in proteins I. A theoretical analysis. J Mol Biol. 1994;236:1369–81.

Novotný J, Bruccoleri RE, Newell J. Twisted hyperboloid (strophoid) as a model of beta-barrels in proteins. J Mol Biol. 1984; 177:567–73.

Pan X, Kortemme T. Recent advances in de novo protein design: principles, methods, and applications. J Biol Chem. 2021;296: 100558.

Rocklin GJ, Chidyausiku TM, Goreshnik I, Ford A, Houliston S, Lemak A, et al. Global analysis of protein folding using massively parallel design, synthesis, and testing. Science. 2017;357: 168–75.

Scott AJ, Niitsu A, Kratochvil HT, Lang EJM, Sengel JT, Dawson WM, et al. Constructing ion channels from water-soluble α-helical barrels. Nat Chem. 2021;13:643–50.

Verkuil R, Kabeli O, Du Y, Wicky BIM, Milles LF, Dauparas J, et al. Language models generalize beyond natural proteins. bioRxiv. 2022. https://doi.org/10.1101/2022.12.21.521521v1

Vorobieva AA. Principles and methods in computational membrane protein design. J Mol Biol. 2021;433:167154.

Vorobieva AA, White P, Liang B, Horne JE, Bera AK, Chow CM, et al. De novo design of transmembrane β barrels. Science. 2021;371: eabc8182.

Wang S, Li W, Liu S, Xu J. RaptorX-property: a web server for protein structure property prediction. Nucleic Acids Res. 2016;44: W430–5.

Wang J, Lisanza S, Juergens D, Tisher D, Watson JL, Castro KM, et al. Scaffolding protein functional sites using deep learning. Science. 2022;377:387–94.

Watson JL, Juergens D, Bennett NR, Trippe BL, Yim J, Eisenach HE, et al. *De novo* design of protein structure and function with RFdiffusion. Nature. 2023;620:1089–100. https://doi.org/10.1038/s41586-023-06415-8

White SH. Biophysical dissection of membrane proteins. Nature. 2009;459:344–6.

Wicky BIM, Milles LF, Courbet A, Ragotte RJ, Dauparas J, Kinfu E, et al. Hallucinating symmetric protein assemblies. Science. 2022;378:56–61.

Wu R, Ding F, Wang R, Shen R, Zhang X, Luo S, et al. High-resolution de novo structure prediction from primary sequence. bioRxiv. 2022. https://doi.org/10.1101/2022.07.21.500999v1

Zhang Y, Skolnick J. TM-align: a protein structure alignment algorithm based on the TM-score. Nucleic Acids Res. 2005;33:2302–9.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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